1. Union-Find Data Structure

(a)

Parent array after each operation (parent$[i] = i$ means i is a root):

Operation	1	2	3	4	5	6	7
Init	1	2	3	4	5	6	7
$\textsc{Union}(1,3)$	1	2	1	4	5	6	7
$\textsc{Union}(1,7)$	1	2	1	4	5	6	1
$\textsc{Union}(2,5)$	1	2	1	4	2	6	1
$\textsc{Union}(4,6)$	1	2	1	4	2	4	1
$\textsc{Union}(2,3)$	1	1	1	4	2	4	1

$\textsc{Union}$(1,3): $|T_1| = |T_3| = 1$ (tie), $3 \to 1$. $\textsc{Union}$(1,7): $|T_7| = 1 < 2 = |T_1|$, $7 \to 1$.

$\textsc{Union}$(2,5): $|T_2| = |T_5| = 1$ (tie), $5 \to 2$. $\textsc{Union}$(4,6): $|T_4| = |T_6| = 1$ (tie), $6 \to 4$.

$\textsc{Union}$(2,3): $\textsc{Find}(2) = 2$, $\textsc{Find}(3) = 1$. $|T_2| = 2 < 3 = |T_1|$, $2 \to 1$.

$\textsc{Find}(5)$: $5 \to 2 \to 1$.

$$\boxed{\textsc{Find}(5) = 1}$$

(b)

Claim: A union-by-size tree of height t has $\geq 2^t$ vertices.

Base ($t = 0$): Height 0 $\implies$ single vertex. $|T| = 1 \geq 2^0$. ✓

Inductive step: Assume for all $t' < t$. Let T be a tree whose height first reaches t via $\textsc{Union}(T_s, T_l)$ with $|T_s| \leq |T_l|$ (root of T_s attached to root of T_l).

For the height to increase: height$(T_s) + 1 = t$, so height$(T_s) = t - 1$.

$$\text{By Inductive Hypothesis:} \quad |T_s| \geq 2^{t-1}$$
$$|T_l| \geq |T_s| \implies |T_l| \geq 2^{t-1}$$
$$|T| = |T_s| + |T_l| \geq 2^{t-1} + 2^{t-1} = 2^t \qquad \square$$

$$\boxed{\text{height}(T) \leq \lfloor \log_2 |T| \rfloor}$$

(c)

Without union-by-size, the direction of each union is arbitrary. We build a chain:

Step	Operation	Direction chosen
1	$\textsc{Union}(1,2)$	$1 \to 2$
2	$\textsc{Union}(3,2)$	$2 \to 3$
3	$\textsc{Union}(4,3)$	$3 \to 4$
4	$\textsc{Union}(5,4)$	$4 \to 5$
5	$\textsc{Union}(6,5)$	$5 \to 6$
6	$\textsc{Union}(7,6)$	$7 \to 6$

Final tree (root $= 6$): path $6 \to 5 \to 4 \to 3 \to 2 \to 1$, with 7 as child of 6.

$$\boxed{\text{height} = 5}$$

2. Dijkstra's Algorithm

(a)

Directed graph $G = (V, E)$, $|V| = 12$, $|E| = 24$, source $s = 1$.

edge	w	edge	w	edge	w
$1 \to 2$	2	$4 \to 5$	1	$7 \to 10$	1
$1 \to 3$	5	$4 \to 7$	5	$7 \to 11$	3
$1 \to 4$	4	$4 \to 8$	3	$8 \to 9$	2
$2 \to 3$	1	$5 \to 6$	4	$8 \to 12$	5
$2 \to 4$	3	$5 \to 9$	2	$9 \to 10$	4
$2 \to 5$	3	$6 \to 7$	1	$9 \to 12$	3
$3 \to 6$	5	$6 \to 10$	5	$10 \to 11$	2
$3 \to 7$	2			$10 \to 12$	1
				$11 \to 12$	4

Dijkstra trace (each row: extract min-distance unvisited vertex, relax outgoing edges):

Iter	Settle u ($d[u]$)	Updates
1	1 (0)	$d[2]{=}2,\ d[3]{=}5,\ d[4]{=}4$
2	2 (2)	$d[3]{=}3,\ d[5]{=}5$
3	3 (3)	$d[6]{=}8,\ d[7]{=}5$
4	4 (4)	$d[8]{=}7$
5	5 (5)	$d[9]{=}7$
6	7 (5)	$d[10]{=}6,\ d[11]{=}8$
7	10 (6)	$d[12]{=}7$
8–12	$8, 9, 12, 6, 11$	(no updates)

Final distances and shortest-path tree:

v	1	2	3	4	5	6	7	8	9	10	11	12
$d[v]$	0	2	3	4	5	8	5	7	7	6	8	7
$\pi[v]$	–	1	2	1	2	3	3	4	5	7	7	10

(b)

Same edges, now undirected. Prim's algorithm from $s = 1$:

Iter	Add v (key)	MST edge	Key updates
1	1 (0)	–	$k[2]{=}2,\ k[3]{=}5,\ k[4]{=}4$
2	2 (2)	$\{1, 2\}$	$k[3]{=}1,\ k[4]{=}3,\ k[5]{=}3$
3	3 (1)	$\{2, 3\}$	$k[6]{=}5,\ k[7]{=}2$
4	7 (2)	$\{3, 7\}$	$k[6]{=}1,\ k[10]{=}1,\ k[11]{=}3$
5	6 (1)	$\{7, 6\}$	–
6	10 (1)	$\{7, 10\}$	$k[9]{=}4,\ k[11]{=}2,\ k[12]{=}1$
7	12 (1)	$\{10, 12\}$	$k[8]{=}5,\ k[9]{=}3$
8	11 (2)	$\{10, 11\}$	–
9	4 (3)	$\{2, 4\}$	$k[5]{=}1,\ k[8]{=}3$
10	5 (1)	$\{4, 5\}$	$k[9]{=}2$
11	9 (2)	$\{5, 9\}$	$k[8]{=}2$
12	8 (2)	$\{9, 8\}$	–

MST weight $= 2 + 1 + 2 + 1 + 1 + 1 + 2 + 3 + 1 + 2 + 2 = 18$.

$$\boxed{W(\text{MST}) = 18}$$

(c)

Directed graph with negative edge weight where Dijkstra fails.

$V = \{1, \ldots, 6\}$, $s = 1$. Edges:

edge	$1 \to 2$	$1 \to 3$	$3 \to 2$	$2 \to 4$	$3 \to 5$	$4 \to 6$	$5 \to 6$
w	1	5	-10	3	2	1	4

Dijkstra settles vertex 2 at $d[2] = 1$ (iter 2). When vertex 3 is later settled (iter 4), the edge $3 \to 2$ with $w = -10$ gives $d[3] + w = 5 + (-10) = -5 < 1 = d[2]$, but vertex 2 is already finalized.

v	1	2	3	4	5	6
Dijkstra $d[v]$	0	1	5	4	7	5
Correct $d^*[v]$	0	-5	5	-2	7	-1

Correct path to 2: $1 \to 3 \to 2$, weight $5 + (-10) = -5$. Dijkstra returns $d[2] = 1 \neq -5$.

$$\boxed{\text{Dijkstra fails on vertices } 2, 4, 6}$$

3. Bellman-Ford Algorithm

(a)

Suppose G contains a negative cycle $C = v_1 \to v_2 \to \cdots \to v_k \to v_1$ reachable from s.

Since C is reachable, after $n-1$ rounds of Bellman-Ford, $d(v_i) < \infty$ for each $v_i \in C$.

Suppose for contradiction $d(u) + w(u,v) \geq d(v)$ for every edge $(u,v) \in E$. Sum over all edges of C:

$$\sum_{i=1}^{k} \big[d(v_i) + w(v_i, v_{i+1})\big] \geq \sum_{i=1}^{k} d(v_{i+1})$$

Since $v_{k+1} = v_1$, the multisets $\{v_1, \ldots, v_k\}$ and $\{v_2, \ldots, v_{k+1}\}$ are identical. The d-value sums cancel:

$$\sum_{i=1}^{k} w(v_i, v_{i+1}) \geq 0$$

This contradicts C being a negative cycle. $\square$

(b)

$V = \{1, \ldots, 6\}$, $s = 1$, 10 directed edges:

edge	1→2	2→3	3→4	4→5	5→6	1→3	1→4	2→4	3→5	2→6
w	1	1	1	1	-5	10	10	10	10	10

Shortest path to vertex 6: $1 \to 2 \to 3 \to 4 \to 5 \to 6$, weight $1{+}1{+}1{+}1{+}(-5) = -1$, using $5 = n-1$ edges.

Trace with worst-case edge ordering (5→6, 4→5, 3→4, 2→3, 1→2, then remaining):

Round	$d[1]$	$d[2]$	$d[3]$	$d[4]$	$d[5]$	$d[6]$
0	0	∞	∞	∞	∞	∞
1	0	1	10	10	20	11
2	0	1	2	10	11	11
3	0	1	2	3	11	6
4	0	1	2	3	4	6
5	0	1	2	3	4	-1

After 4 rounds: $d[6] = 6$. After 5 rounds: $d[6] = -1$.

$$\boxed{d[6] = 6 \neq -1 = d^*(6) \text{ after 4 rounds}}$$